

Experiment 18

Understand the KNN algorithm for the given data set.

Program:

```
import numpy as np
import pandas as pd

df=pd.read_csv('Iris.csv')
df.info()
df.variety.value_counts()
df.head()

features=df.iloc[:, :-1].values
label=df.iloc[:, 4].values

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split(
    features, label, test_size=.2, random_state=1
)

model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)

print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))

from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))

from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

Output

Output:
Scores:
0.9583333333333334
1.0

Confusion Matrix:
[[50 0 0]
[0 47 3]
[0 2 48]]

Classification Report:

	precision	recall	f1-score	support
Setosa	1.00	1.00	1.00	50
Versicolor	0.96	0.94	0.95	50
Virginica	0.94	0.96	0.95	50
accuracy			0.97	150
macro avg	0.97	0.97	0.97	150
weighted avg	0.97	0.97	0.97	150